Multi-Material Fabrication for Biodegradable Structures

Enabling the printing of porous mycelium composite structures

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Our awareness of the earth's depleting resources has directed focus towards biomaterials, which can be extracted sustainably and biodegraded after use. Current fabrication of biomaterial structures is still restricted in strength and geometry, limiting its use in construction. This paper presents a novel two-phase multi-material fabrication process to create mycelium composite structures of higher porosity and complexity with speculated improvements in strength. First, cellulose pulp inoculated with mycelium is extruded. Then, each layer is filled by a secondary supporting material. This material, in the form of a gravel- and sand-slurry, acts as an inhospitable medium steering mycelial growth, additionally improving aeration to produce stronger structures. After an intermediate growth period, the secondary material, reusable in a closed-loop production model, is removed to reveal the fully-grown mycelium structure. The paper reports on each of the three aspects: the fabrication process, material experimentation of primary and secondary substrates, as well as geometry of varying porosity and performance.

Keywords: *biomaterials, mycelium, biodegradable structures, robotic fabrication, additive manufacturing*

INTRODUCTION

The global construction industry is on an unsustainable path of consumption, with raw materials usage expecting to double in 2060 (OECD, 2018). As one of the largest consumers of raw materials and contributors of carbon emission (OECD, 2019), there is an urgent need to reconsider the way we design and construct our built environment. Currently, there is a mismatch between material durability and usage duration. Materials such as concrete can be made extremely durable - albeit accumulating high maintenance costs (Saad, 2016) - such that the end of a building's life depends on its use becoming obsolete rather than the aging of the material. In light of the new ecological age, this is a paradigm worth reconsidering. While materials should be strong and sturdy to be functional, it should not be so that it remains in the environment when it is no longer needed.

Biomaterials are a potential alternative that can be manufactured sustainably and returned to nature after completing its purpose. Mycelium, in particular, is an organic polymeric composite material com-

prising chitin, cellulose, and proteins. Its self-growing nature constitutes a natural manufacturing process that requires minimal additional energy. As a material, it has yet to reach sufficient mechanical strength (Islam et al., 2017) for construction applications in terms of its load-bearing properties. Industry use of mycelium involves casting of mycelium into brick moulds, and additionally, compression of the bricks to improve performance. However, this limits the geometry achievable, as well as results in varying mechanical properties across the brick due to the unequal access to oxygen which determines the level of mycelium colonisation. While more complex forms can be achieved with 3D printing of mycelium, it is still restricted in geometry due to layer constraints in printing biomaterials.

Adapting solutions from the field of 3D printing, our contribution lies in a two-phase fabrication process that includes the addition of a secondary material serving as a support structure. This removes the geometrical constraints of printing and allows for the fabricable form to be growth-optimised rather than dictated by the printing method. The printing takes place in an open container, the size of which can be decided by the bounding volume of the desired print. After the extrusion of each inoculated pulp layer, the remaining negative space is filled with a secondary material to create a supporting bed for the next layer to be printed on. Our contribution lies in the novel fabrication process for robotically steering the extrusion of mycelium (Figure 1). The method extends the geometrical and fabrication possibilities of biomaterials by removing the limitations of typical bio-printing, improving potential for architectural applications. Furthermore, the increased possibility in complexity and thus porosity allows for the design of forms that are better optimised for the growth of the living organism. The focus of this investigation is to develop a workflow steering three key aspects: fabrication, material, and geometry.

BACKGROUND

On Biomaterials and Mycelium

Abundant in nature with the ability to be fabricated in low-energy processes (Wijk et al., 2015), biomaterials present an opportunity for sustainable construction. Structures such as the Aguahoja from Mediated Matter Group demonstrate the potential of creating bio-composites with high material tunability. By altering the constituent composition, the degree of stiffness, opacity, and decay can be controlled. With biomaterials, the structure can also be returned to nature after it has fulfilled its purpose, creating a bio-based circular design process. Its degradation and susceptibility to changes in the environment defy the current building paradigm where objects are built to last. Rather than viewing its decay as an obstacle, it

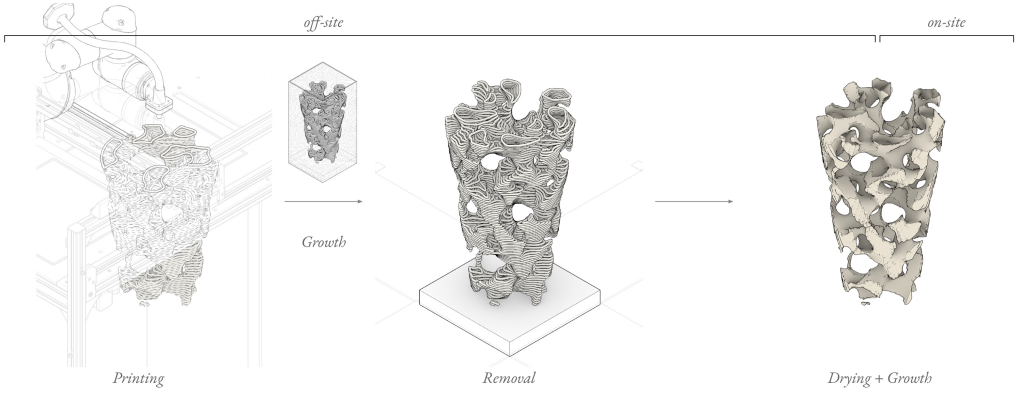


Figure 1
Overall process of printing, growing and drying

could instead give rise to a new design approach that accounts for the different stages across a structure's lifespan.

Mycelium presents interesting alternatives for rethinking material performance and expectancy. As a living organism, it reacts to external stimuli, restructuring its hyphal network to optimise nutrient intake. As a material, it morphs continually to generate varying properties across the mycelial body. It self-replicates to create seemingly endless material in the presence of suitable substrates that can be fully sourced from waste (Jambaro et al., 2014). When subjected to trauma, it is a 'self-healing' material capable of sealing breaks in the structure. Its advantages in sound, fire insulation, and water-proofing (Jones et al., 2018) present interesting application potential. Furthermore, the abundance of fungi species is largely unexplored in their application in construction materials and future research could yield strains and composites of improved performance. The ease of replicability, low cost and environmental sustainability indicate its potential to play a significant role in the future sustainable construction.

Fabrication with Mycelium

Fabrication of mycelium for industrial use is still undergoing development. The main process explored to date is the casting and compression of mycelium bricks (Ross, 2016) - improving strength by maximising its density. An efficient and practical way of using mycelium as a material, this technique is being used to produce insulation panels by firms such as Mogu (Appels et al., 2019). In terms of its use as a construction material, mycelium bricks and panels have been used in temporary structures such as the Hy-Fi tower (Nagy et al., 2015), the Growing Pavilion (Elsacker et al., 2021), and the works of Philip Ross. However, the strength of mycelium composites falls far below concrete even after compression (Islam et al., 2017), which is currently the biggest obstacle in its use in construction. Compressive strength is greatly dependent on the substrate constituents (Vidholdova et al. 2019), and there is potential for stronger composites as new fungal strains and substrate combinations are tested for construction applications. Stronger forms could potentially be generated through design of forms better optimised for the colonisation of mycelium. Extent of mycelium colonisation is strongly related to the access to oxygen (Tang et al., 2003), hence cast bricks in solid volume result in lower colonisation in the overall volume and limit potential strength of the form. By designing for forms with greater surface area, composite strength could be improved as mycelium coverage increases. In nature, termite-constructed fungus nests are also made sufficiently porous to allow for gaseous exchange to keep both termites and fungus alive (Turner, 2005). These natural forms, optimised for fungi growth with a constant width, also demonstrate a high level of complexity that can be achieved with mycelial composites.

Projects such as the Mycotree (Heisel et al., 2018) show the possibility of mycelium being used structurally by designing forms that optimise material properties. Besides casting, more complex geometry can be achieved with 3D printing as demonstrated by several prototypes by Blast Studio and Bio-Digital Matter Lab. These forms greatly improve the aeration of the structure through increased surface area and reduced mycelial wall width. However, the geometrical possibilities are limited by the restrictions of 3D printing: sagging of layers due to excessive self-weight and failed overhangs that exceed the maximum possible angle (Figure 2).

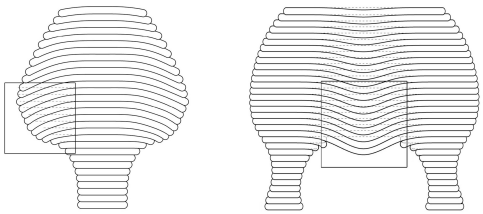


Figure 2
Sagging due to self weight (left), Failing overhang (right)

This research aims to increase the geometrical possibilities of mycelium composites and overcome the limitations of mycelium fabrication to create

biodegradable structures with wider architectural applications. The experiments reported in this paper aim to achieve the following goals:

- Fabrication: Minimal waste production, enabling production of complex geometries, promoting growth of mycelium
- Design: Improved myceliation of structure, measurable porosity
- Materials: Locally sourced waste substrate and reusable secondary material

FABRICATION METHOD

A two-phase multi-material fabrication process is explored in this paper to create mycelium composite structures of higher porosity and complexity. These forms not only test the limit of geometrical complexity, but also present an opportunity to better fabricate with mycelium by catering to its biological properties and providing better aeration. To make these intricate structures possible, the proposed method overcomes current printing limitations by introducing a secondary material to serve as structural support. A fabrication set-up is designed to both extrude the primary inoculated pulp which makes up the form, as well as deposit the secondary material that supports and shapes it.

Fabrication Set-up

The printing process takes place within an open container which is sealed post-fabrication with a microfilter lid to minimise exposure to contaminants during the growth process. The overall set-up (Figure 3) designed for this method comprises the extrusion set-up for the inoculated pulp, which is controlled by a collaborative robot affixed with a customized extrusion system, and a deposition set-up which enables the secondary material to be applied in the remaining negative area.

The extrusion set-up comprises a collaborative robot, nozzle, connecting pipe, storage unit to hold the pulp and pulp-pushing device. The latter can take the form of either an air compressor and pneu-

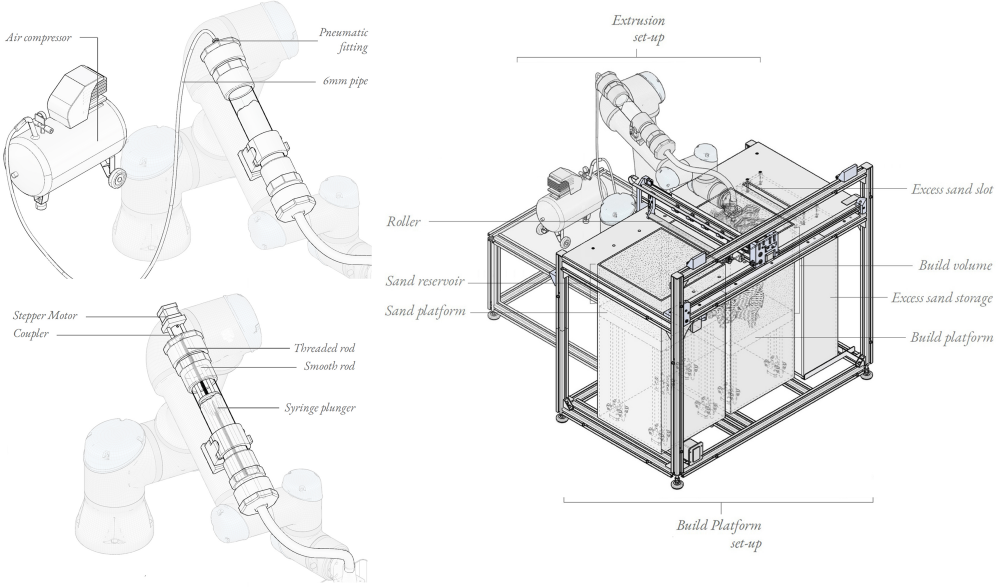


Figure 3
Designed set-up: Pneumatic extrusion set-up (top left), Mechanical extrusion set-up (bottom left) Overall set-up: Extrusion + Deposition (right)

Figure 4
Set-up prototype, using a UR5 robot



matic fitting set-up or a stepper motor and plunger set-up; both systems have been proven to work in projects with similar wet pulp extrusion (Wang et al., 2018). A pneumatic set-up was used for this prototype for its simpler set-up (Figure 4).

The deposition set-up can be partially adapted from a Selective Laser Sintering (SLS) machine (Hussain et al., 2019), where two moving platforms for the secondary material and printed artefact move accordingly each layer. A roller pushes the secondary material from the top of the reservoir to fill in the negative space in the build volume; any excess is pushed through the slot where it can later be reused. An alternative set-up would be the design of an aggregate deposition device (beyond the scope of this paper) to deposit the material at negative areas. This area of research in aggregate deposition is to be further explored beyond this paper. A manual deposition method is used for this set of experiments.

Fabrication Process

The fabrication process, the first step in the overall timeline (Figure 1), can be divided into two phases. In phase one, inoculated pulp is printed with a collaborative robot arm. In phase two, the remaining negative space left over in each layer is filled by a secondary supporting material, completing the layer. This repeats for each layer until the structure is fully printed, and the container is sealed with a microfilter lid as the artefact enters the first growth phase. This technique leverages on the delayed forming process of mycelium, which takes time to colonise the substrate and to create a rigid structure. After a period of about 14 days, the structure, along with the adjacent secondary material (which is reusable in a closed-loop production model) can be removed from the container (Figure 5). At this point, a decision can be made whether to integrate the secondary material into the form as a texture or removed completely. After the removal and exposure to the external environment, the structure undergoes an intermediate growth and drying phase where it begins solidifying into the final form (Figure 6). Any remaining secondary material that has not been integrated into the mycelium starts to fall from the structure from this point in time.



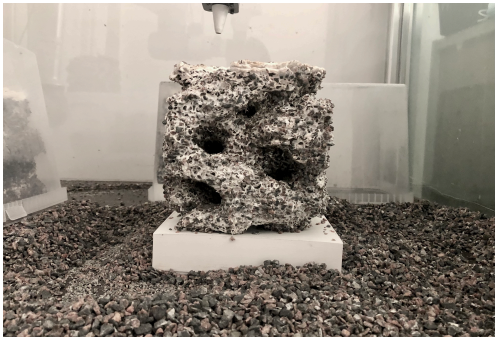


Figure 5
Grown artefact after removal from container

Figure 6
Unveiled artefact after secondary growth and removing of secondary material

EXPERIMENTS

A series of experiments were carried out at different points to test the workability of the fabrication technique and remedy any potential issues, as well as prove the feasibility of shaping mycelium composites with a secondary material during its growth phase. Introduction of contaminants was minimised by sterilising the substrates (prior to inoculation) and secondary material with a pressure cooker, reducing contact time of the biocomposite materials to the external environment, and constant cleaning of the set-up with alcohol. This proved sufficient for the growth of mycelium in this set-up and no visible contamination was observed in the final experiments. This research uses Ganoderma Lucidum (G. lucidum) for its fast-growing nature, structural performance and extensive published use in research. However, this is interchangeable with other species in the application of this technique, especially in a site-specific context in consideration of biodiversity compatibility.

Fabrication Set-up Prototype

The pneumatic extrusion set-up (Figure 4) was built to investigate the steering of the primary substrate using a robot arm. For initial testing purposes, the secondary material deposit was done manually by pouring it into the remaining unprinted space each layer. The variables in this set-up include the size of the print storage box, which restricts the printable extents, as well as nozzle size which affects the resolution of the print. Several iterations of the set-up were visited, each improving upon the main issue arising from clogging of the pulp along the nozzle. The recipe of the extrusion pulp was found to be crucial in the success of the extrusion, and should be thoroughly blended and filtered to ensure minimal clogging. This set-up was used in a series of material experiments with recipes varying primary substrate composition and secondary material type to determine the best mix for growth of the artefact.

Primary Substrate (Extrusion Pulp)

For a suitable extrusion pulp, there are two main criteria to create the primary substrate: suitability for fabrication and support of fungus growth. The main composition was experimented with varying concentrations of mycelium spawn, mycelium feed, binding agent and water. In alignment with the experiment's aim of using biodegradable materials, the mycelium feed would ideally be sourced from a waste stream that can be broken down by the fungus, such as non-recyclable paper including food-soiled paper packaging and wax/plastic-lined paper. For the context of this research, waste paper pulp has been selected for its high cellulose content and ease of availability. Xanthan gum was selected from a range of binding agents for its performance in holding the substrate together and keeping it suitably viscous. The water content in the mixture, a necessity for mycelial growth, had to be sufficient without causing excessive pooling within the container such that the mycelium drowns. However, this can also be mitigated by elevating the print with additional layers of secondary material prior to printing. The final com-

position used in successful subsequent experiments comprises, by weight: 11% mycelium spawn, 56% paper pulp, 1% xanthan gum and 32% water by weight. This achieved a smooth extrusion process with viscosity low enough to flow through the nozzle yet sufficiently solid for shape retention during and after the printing process. The recipe allowed for sufficient mycelium colonisation of the printed structure (Figure 5) within 14 days.

Secondary Material

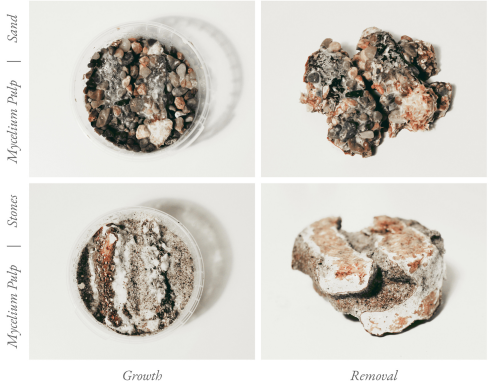
The inorganic nature of the secondary material allows it to act as an inhospitable medium steering mycelial growth and supporting the weight of the extrusion pulp. The scale of the secondary material also affects the forming of mycelial skin on the artefact. Larger aggregates are able to provide greater aeration, but reduce the resolution of the final artefact as its removal leaves an imprint on the surface (Figure 5). Hence, the coarseness of the surface can be controlled by varying the diameter of the secondary material. Depending on the application, a decision to retain the secondary material on the structure after the growth phase can also be made. Besides the appearance in texture, this could have implications on the structural performance of the final artefact, possibly improving stiffness and load-bearing properties although this is not explored in this paper.

A series of experiments (Figure 7) were conducted to investigate the impact of choice of secondary materials. Here, sand (0.02-0.05mm) and gravel (4-8mm) were used for their contrasting scales, left adjacent to a mix of inoculated wood chip substrate in an enclosed container with a micro-filter. After 7 days, the contents were removed and left to dry for a day. The results give an insight to the interaction between primary and secondary material during the growth period. A gradient of bonding corresponding to the extent of mycelium colonisation is observed at the boundary between the two materials, as the secondary material gets increasingly difficult to remove without destroying the substrate. Experiments attempting to shape form with the secondary material (Figure 8) by manually extruding orthogonal stripes with a piping bag proved relatively successful. The effect of the secondary material was clearly observable in the grown artefacts, with the sand and gravel creating vastly different textures. It is observed that the sand left a much finer texture along the boundary and could be easily removed with water. Overall, the selection of secondary material results in varying results of surface texture, and possibly, strength of the final artefact.

Figure 7
Series of material experiments with sand, gravel and both



Figure 8
Series of experiments exploring possibility of forming with the secondary material



In order to test the proposed fabrication process, a sufficiently complex form pushing the typical sagging and overhang limits should be attempted. Minimal surfaces are engaged in the design strategy for



Scale: 0.2
Volume of mycelium: 55.0%
Area of mycelium: 202.0%

Scale: 0.3
Volume of mycelium: 55.0%
Area of mycelium: 152.3%

Scale: 0.4
Volume of mycelium: 54.2%
Area of mycelium: 123.6%

Scale: 0.25
Volume of mycelium: 55.0%
Area of mycelium: 240.0%

Scale: 0.45
Volume of mycelium: 55.1%
Area of mycelium: 145.1%

Figure 9
Series of porosity explorations with minimal surfaces

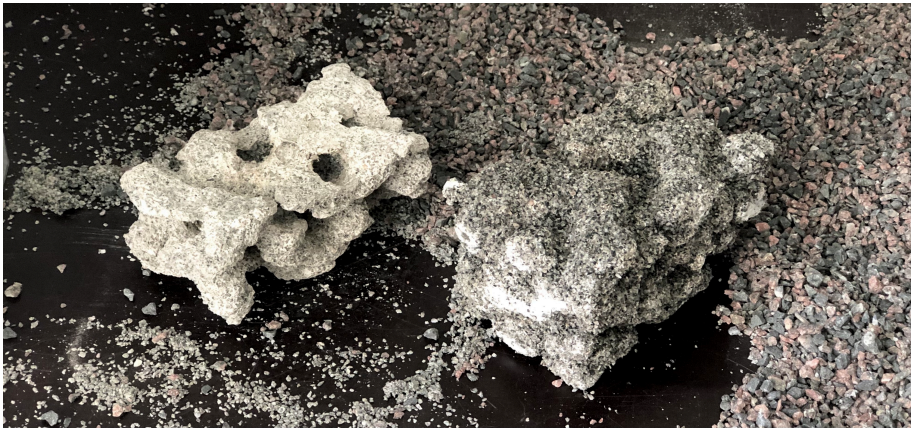


Figure 10
Initial prototypes in mould of dimension 150x90x90mm, varying porosity



Figure 11
Scaled-up prototypes in mould of dimension 150x150x150mm, varying porosity

its ability to generate porosity within a given volume, increasing the surface area for improved aeration for the mycelium. Porosity is compared by taking the difference in volume between the final mininal surface form and the boundary volume (Figure 9). Its overhanging volumes also test the ability of the proposed multi-material fabrication system, while allowing for the removal of secondary material after the growth period due to the interconnected volume. Porosity and thickness of mycelial composite walls can be controlled to produce varying material properties of artefacts. Performance studies of gyroid forms with other materials have proven to have improved deformation (Yang, 2019) which could be transferable to bio-composite forms as well. Results in Figure 10 were prototyped in the Fisherkoch form in a mould of dimensions 150x90x90mm. However, the scale of the print proved to be too small in comparison to the thickness of the extrusion thickness (5mm) hence the details of the form were less discernible. This was remedied in another set of prints scaling up the size of artefacts with a larger cubic mould of width 150mm (Figure 11).

CONCLUSIONS

The experiments demonstrate the potential of a multi-material process of fabricating with mycelium bio-composites. While the fabrication method could be further refined, the prototypes have shed some light on potential pitfalls and areas of improvement such as the relationship between scale of material, extrusion thickness and print dimension that would affect print resolution. We pinpointed a crucial problem linked to the clogging of pulp that emphasised the need for a good water-to-binding agent ratio and sufficient blending of pulp. With a finer pulp, nozzle size can also be reduced to create finer prints. The process of manually depositing the aggregates also highlighted the importance of precision in the secondary material deposition set-up as significant effort had to be exerted to prevent the aggregates from falling and sticking onto the extrusion pulp. Further considerations will also include the integration of secondary material into the primary pulp itself to test the effect on structural performance.

The primary material explorations provide an adequate extrusion pulp for the proposed method, with healthy mycelial growth and retention of shape. Introduction of the secondary material not only opens the possibility of more complex geometry, but also brings in aspects of porosity and aeration that could improve the final structural performance and growth of the mycelial skin. Furthermore, the choice of secondary material adds an interesting design aspect in terms of texture, whether by fusing it with the mycelial skin or removing it to reveal the indents left behind. An extension of this research in the material aspect would include further exploration of growing inorganic material with the living matter, as well as simulating the interaction and evolution of the two contrasting elements.

PERSPECTIVES

Further investigation can be conducted to observe the evolution of the grown structure. By working with minimal surfaces, a clearer relationship between porosity, surface area and wall thickness can be established with more results. The porosity of the artefacts and thickness of composite walls will likely correlate with the rate of degradation, a factor which can be controlled in the design process. This would allow designers to programme the lifespan of the structure, planning the various stages of degradation according to the desired function. Appearance of the skin can be varied by choosing a keep the secondary material and choosing a suitable scale of the aggregates. By keeping the secondary material, the evolution of the artefact also includes the eventual detachment of secondary material. A simulation could be developed to better understand and work with these evolving factors in a biodegradable structure. Depending on the site context, considerations in fungal strain and substrate constituents have to be considered and determined to be compatible with existing biodiversity especially if introduced in a large quantity. The architectural applications of this fab-

rication process could involve a shifting programme that evolves along with the structure. Overall, this fabrication process expands the possibilities of fabricating with bio-composites in terms of geometry, performance and lifespan.

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